

FTP-OPS-SOP-014 — Lockout-Tagout (LOTO) and Isolation Procedure for Rotating Equipment

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Department	Operations	Document Owner	Plant Manager, Operations
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Applicable Units	Utilities Block, Rotating Equipment Bay, Compressor House	Classification	Internal Use

Revision History

Rev	Date	Description of Change	Prepared By	Approved By
0.0	22-Jan-2023	Initial issue	H. Deshmukh, Sr. Engineer (Ops)	Plant Manager
1.0	15-Jun-2024	Added isolation verification checklist; updated PPE table	S. Kulkarni, Safety Officer	Plant Manager
2.0	10-Jul-2026	Revised group-lock box procedure; added multi-agent asset tagging cross-reference; updated emergency contact matrix	S. Kulkarni, Safety Officer	Plant Manager

1. Scope

This procedure defines the mandatory requirements for isolation, lockout, and tagout (LOTO) of rotating and static equipment prior to maintenance, inspection, or repair activities within FortTrace Plant Operations facilities. It applies to all rotating machinery including centrifugal pumps, compressors, blowers, agitators, and their associated drive systems (motors, gearboxes, couplings) located in the Utilities Block, Rotating Equipment Bay, and Compressor House.

This procedure is applicable to all FortTrace employees, contract maintenance personnel, and third-party service engineers performing work on covered equipment. It does not cover electrical high-voltage switchgear isolation, which is addressed separately under FTP-OPS-SOP-009 (Electrical Isolation and Permit-to-Work Procedure), or confined space entry, covered under FTP-OPS-SOP-021.

2. Responsibilities

2.1 Shift In-Charge / Area Owner

Authorizes issuance of the isolation permit, verifies that the equipment identified on the permit matches the physical asset tag, and confirms that the affected process area has been notified before work commences.

2.2 Maintenance Execution Engineer

Raises the isolation request in the Permit-to-Work (PTW) system, prepares the isolation point list for the specific asset, and coordinates with Operations to schedule the isolation window.

2.3 Authorized Isolator

Physically performs the isolation (electrical breaker racking, valve closure, blinding, spring isolation of couplings), applies personal locks and danger tags at each isolation point, and signs the isolation certificate confirming zero-energy state.

2.4 Individual Workers

Apply personal locks to the group lock box before commencing work, verify isolation independently at the job site using a try-out test, and remove personal locks only after confirming completion of assigned work.

2.5 Safety Officer / HSE Department

Conducts periodic audits of active LOTO permits, maintains the isolation register, investigates any near-miss or incident related to isolation failure, and reviews this procedure annually.

3. Procedure

3.1 Pre-Isolation Planning

- 1 Identify the equipment tag number and cross-check against the P&ID; and equipment master list.
- 2 Raise a Permit-to-Work (PTW) request in the plant PTW system at least 4 hours prior to the planned isolation window for routine work, or as per emergency isolation protocol for breakdown work.
- 3 Prepare an isolation point list identifying every energy source: electrical supply, process fluid lines, pneumatic/hydraulic supply, and stored mechanical energy (springs, counterweights, elevated components).
- 4 Verify that a suitable lock, multi-lock hasp, or group lock box is available for the number of workers expected to be involved.

3.2 Isolation Execution

- 1 Notify the control room and affected process area before initiating isolation.
- 2 Shut down the equipment through normal operating procedure; do not isolate a running machine by breaker trip alone.
- 3 De-energize the electrical supply at the motor control center (MCC); rack out the breaker or remove the fuse; apply a personal danger tag and lock at the isolation point.
- 4 Close and lock all associated process isolation valves (suction, discharge, drain) and install line blinds where required by the isolation point list.
- 5 Release stored energy: bleed residual pressure, block any spring-loaded or gravity-loaded components, and allow rotating parts to come to a complete stop.
- 6 Perform a try-out test (attempt to start the equipment locally and from the control room) to confirm zero-energy state before allowing work to proceed.
- 7 Complete and sign the isolation certificate; attach it to the PTW documentation at the job site.

3.3 Group Lockout (Multiple Workers)

Where more than one crew or discipline is working on the same isolated equipment, the Authorized Isolator shall place the master isolation lock on a group lock box. Each worker or crew supervisor then applies an individual personal lock to the group lock box before commencing work. No worker may begin work on the equipment without their own lock on the box, and no lock may be removed by anyone other than the individual who applied it, except under the Lock Removal by Other Than Authorized Person exception in Section 3.5.

3.4 De-Isolation and Restoration

- 1 Confirm all work is complete, tools and temporary blinds are removed, and the area is clear of personnel.
- 2 Each worker removes their individual lock from the group lock box only after confirming their portion of work is complete.
- 3 The Authorized Isolator removes line blinds, reopens process valves in the correct sequence, and restores electrical supply only after all personal locks have been removed.
- 4 Conduct a pre-start check and notify the control room before restoring equipment to service.
- 5 Close out the isolation certificate and archive it with the PTW record.

3.5 Lock Removal by Other Than Authorized Person (Exception)

If the individual who applied a personal lock is unavailable to remove it (e.g., end of shift, leave), the lock may only be removed by the Shift In-Charge after: (a) making all reasonable efforts to contact the individual, (b) verifying that the individual is not present in the isolated area, and (c) documenting the removal in the isolation register with date, time, and reason. The individual must be informed before resuming work on the equipment.

3.6 Required PPE

Task	Minimum PPE
Electrical isolation at MCC/breaker	Arc-rated coveralls, insulated gloves (Class 0), safety glasses, dielectric footwear
Valve isolation / line blinding	Coveralls, chemical splash goggles, nitrile gloves, safety shoes
Working near rotating equipment	Hard hat, safety glasses, hearing protection, close-fitting coveralls (no loose clothing)

4. References

- FTP-OPS-SOP-009 — Electrical Isolation and Permit-to-Work Procedure
- FTP-OPS-SOP-021 — Confined Space Entry Procedure
- FTP-HSE-STD-002 — Personal Protective Equipment Standard
- OISD-STD-105 — Work Permit Systems (reference standard, industry practice)

5. Approval

Prepared By	Reviewed By	Approved By
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